Topic 2.2 - Change in Linear and Exponential Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ______________ Class: ______________ Date: ______________

Why this topic matters

Linear and exponential functions can both increase or decrease, but they organize change differently. A linear model adds the same amount over equal input intervals; an exponential model multiplies by the same factor. Strong comparisons focus on the invariant and on what the parameters mean in context.

Learning targets

- Distinguish additive change from proportional change across tables, graphs, formulas, and contexts.
- Construct linear and exponential functions from two values or from stated rates.
- Compare models, interpret intersections, and respect discrete versus continuous domains.

Language and structure

Term	Meaning in this topic
additive change	Equal input intervals produce equal output differences.
proportional change	Equal input intervals produce equal output ratios or percent changes.
initial value	The function value at the chosen reference input, often $x = 0$.
growth factor	The multiplier b in ab^x; the percent change is $(b-1)100$%.
intersection	An input-output pair satisfying both models; in context it marks equal predicted values.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A school campaign can gain supporters in two ways. Plan L starts with 80 supporters and adds 26 each week. Plan E starts with 80 supporters and grows by 22% each week. Which plan is larger during the first month, and which is likely larger later? State what must be calculated rather than guessed.

Concept 1: Recognizing the invariant

Core idea

Linear change has constant differences; exponential change has constant ratios for equal input intervals.

- Linear form: $L(x) = mx + b$, where m is change in output per one input unit.
- Exponential form: $E(x) = ab^x$, where b is the factor per one input unit.
- A graph's shape alone is not enough; use coordinates, scale, or a defining rule.

Worked example - annotate the reasoning

Problem. A table contains $(0, 18), (2, 30), (4, 42), (6, 54)$. Determine whether a linear or exponential model is appropriate and write the model.
Model reasoning. Every increase of 2 in x adds 12 to y, so the rate is 6 per input unit. The model is $y = 18 + 6x$.

Check your understanding

Independent

The values of y at $x = 0, 1, 2, 3$ are $5, 7.5, 11.25, 16.875$. Identify the model family and write a rule.

Concept 2: Building models from two values

Core idea

For a linear model use slope; for an exponential model use a ratio adjusted for the input gap.

- Linear: $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, then solve for b.
- Exponential: $\dfrac{y_2}{y_1} = b^{x_2 - x_1}$, then solve for a.
- Units belong to m; the growth factor b is dimensionless but tied to a time interval.

Worked example - annotate the reasoning

Problem. Construct a linear function through $(3, 17)$ and $(9, 47)$, and an exponential function through $(1, 48)$ and $(4, 384)$.
Model reasoning. Linear: $m = (47 - 17)/(9 - 3) = 5$, so $L(x) = 5x + 2$. Exponential: $384/48 = 8 = b^3$, so $b = 2$ and $48 = a(2)$, giving $E(x) = 24(2)^x$.

Check your understanding

Independent

An exponential function satisfies $f(2) = 75$ and $f(6) = 1200$. Find $f(x)$.

Concept 3: Comparing models and interpreting intersections

Core idea

The model with the larger current value need not have the larger long-run value.

- Compare functions at the same inputs and identify the first input at which the ordering changes.
- An intersection means equal model predictions, not that the real quantities must be equal.
- Extrapolation is only defensible while the assumptions and context remain reasonable.

Worked example - annotate the reasoning

Problem. Plans A and B cost $A(t) = 120 + 35t$ and $B(t) = 120(1.18)^t$ QAR after t months. Find the first whole month when B costs more than A.
Model reasoning. At $t = 6$, $A = 330$ and $B \approx 323.95$. At $t = 7$, $A = 365$ and $B \approx 382.26$. Therefore the first whole month is 7.

Check your understanding

Independent

Explain why the statement “exponential always grows faster than linear” is incomplete.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Identify the model family and rule for the table $(0, 240), (1, 204), (2, 173.4), (3, 147.39)$.

Exit ticket 2

No calculator

A linear function passes through $(4, -7)$ and $(10, 20)$. Find its rate of change.

Exit ticket 3

No calculator

Interpret the intersection of two revenue models in context.